

LESSON 4.6

FINDING CIRCUMFERENCE



LESSON INTRODUCTION

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

LEARNING TARGETS

- I can find the circumference of a circle.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- What is the relationship between the diameter and circumference of a circle?
- How do you determine the circumference of a circle?

LESSON NARRATIVE

Students build on their discovery that the ratio of the circumference to the radius is a constant value. Students learn that this relationship can be represented by the symbol of pi. Students derive the circumference formula, and follow this with an exploration in which different representations of the approximation of pi are used to find circumference. Students then apply the formula for the circumference of a circle to solve problems.

COMPONENT SUMMARY

4.6.1 WARM-UP (~5 MIN.)

Explore utility of measurements of real-world circles

4.6.2 GUIDED INSTRUCTION (15 MIN.)

Calculate circumferences using approximations of pi

4.6.3 GUIDED PRACTICE (10 MIN.)

Identify and correct errors in calculating circumference

4.6.4 YOUR TURN! (10 MIN.)

Practice calculating circumferences in real-world contexts

4.6.5 WRAP-UP (~10 MIN.)

Explore which form of an answer yields a more accurate calculation of circumference

RESOURCES

GLOSSARY

Key Terms:

- Pi (π)

Additional Terms:

- N/A

MATERIALS

- Calculators

ADDITIONAL PREPARATION

- Consider modifying the people, places, and things in 4.6.4 Your Turn! to appeal to your students' interests and everyday lives.
- If time is an issue, then consider assigning Your Turn! for homework.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may substitute the radius for the diameter or the diameter for the radius when using the circumference formulas $C = \pi d$ and $C = 2\pi r$.

THINGS TO CONSIDER

- In 4.6.2 Guided Instruction, connect the exploration from Lesson 5 to ensure students make the connection. Use the objects from the exploration to remind them of the relationship they discovered.
- Emphasize the relationship between radius and diameter. Remind students to check their work and make sure they substituted the correct value into the correct formula.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

Display language from student discussions during the exploration in Lesson 5 to refer to during the guided instruction. Help students improve the language based on what students learn through the guided instruction. Then add new terms or phrases to the list to help students build their mathematical language.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.5.1 Patterns and Structure

In 4.6.2 Guided Instruction, students explore, discover, and represent the ratio of the circumference to the diameter as pi. Students develop an understanding of the relationship by looking for and making use of structure.

DIFFERENTIATE INSTRUCTION

GeoGebra offers interactive, digital applets that students can explore while conceptualizing the circumference of a circle.

For example, teachers could project a GeoGebra applet of their choosing for the whole class to launch the Guided Instruction and have students make conjectures about the formula for circumference.

4.6.1 WARM-UP: CIRCULAR OBJECTS

TEACHER MOVES

Think-Pair-Share

Give students 1 minute of quiet think time followed by 2 minutes to discuss their explanations with a partner. Finally, ask a few students to share with the whole class.



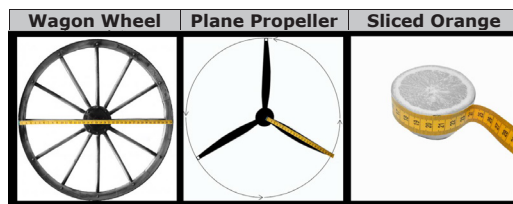
4.6.1 Warm-Up: Circular Objects

Three pictures of circular objects, with measurement tools, are shown. The measurement tool on each picture reads as follows:

Wagon wheel: 3 feet (ft)

Plane propeller: 24 inches (in)

Sliced Orange: 20 centimeters (cm)



- For each object, what does the measurement represent?

Wagon Wheel	Plane Propeller	Sliced Orange
- radius - diameter - circumference	- radius - diameter - circumference	- radius - diameter - circumference

- Choose one of the objects and explain why you think it would be helpful to know the measurement given.
Answers will vary. It will be helpful to know the measurement of the orange for cooking purposes. It will be helpful to know the measurement of the wheel in case you need to replace the wheel.



4.6.2 Guided Instruction: Circumference

The relationship between the diameter, d , and the circumference, C , of a circle is proportional because the ratio of $C:d$ is always the same. This ratio, $\frac{C}{d}$, is called pi. The Greek symbol π is used to represent this value.

1. Rewrite the equation $\pi = \frac{C}{d}$ so that circumference, C , is written in terms of diameter, d .

$$C = \pi d$$

2. Describe the relationship between the radius and the diameter of a circle.

Radius is half of the diameter, or the diameter is two times the radius.

3. Rewrite the equation from question 1 so that circumference, C , is written in terms of radius, r , by substituting how many radii make up the diameter, d .

$$C = 2\pi r$$



Pi (π) is the symbol designating the ratio of the circumference of a circle to its diameter. It is an irrational number.

Common approximations for pi are 3.14, $\frac{22}{7}$ or $\frac{355}{113}$.

The symbol $\approx$ represents "is approximately." It is used when a value has been rounded off or estimated.

The symbol π is used to represent the exact value of pi. Approximations of this value are often used when solving problems since pi is irrational and, therefore, has an infinite number of digits. When 3.14, $\frac{22}{7}$, or $\frac{355}{113}$ are used in place of π to solve problems, the answers are considered approximate values. Using the π button on a calculator will also result in an approximation because the result shown on the calculator does not include all the digits. Also, when using the value on the calculator, it is often rounded.

4. Complete the table to rewrite the circumference formulas using the common approximations for π .

Approximate Value of pi (π)	$C = \pi d$	$C = 2\pi r$
$\pi \approx 3.14$	$C \approx 3.14d$	$C \approx 6.28r$
$\pi \approx \frac{22}{7}$	$C = \frac{22}{7}d$	$C = \frac{44}{7}r$
$\pi \approx \frac{355}{113}$	$C = \frac{355}{113}d$	$C = \frac{710}{113}r$

4.6.2 GUIDED INSTRUCTION:
CIRCUMFERENCE

ENRICHMENT

Notice and Wonder

What do you notice and wonder about the answers to questions 1 and 3? How are they similar? How are they different? What does that make you think about finding the circumference of a circle? Do you need radius or diameter?

TEACHER MOVES

Consider using a calculator to compare the decimal equivalent of the three approximate values of pi to the more precise value of pi returned on a calculator.

4.6.2 GUIDED INSTRUCTION: CIRCUMFERENCE (CONTINUED)

DIFFERENTIATE INSTRUCTION

Consider splitting students into groups and asking each group to use a different approximation for pi in their calculations: 3.14, $\frac{22}{7}$, or $\frac{355}{113}$.

Ask each group to display and explain their work to provide visual and auditory entry points. Conclude with a discussion about which approximation of pi results in the most accurate measurement of the circumference.

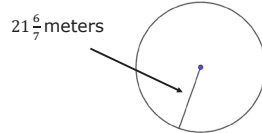
5. Determine the circumference of each circle, both in terms of π and the approximation. For approximate values, use the approximation listed. Lengths are given in centimeters (cm). The circles are not drawn to scale.

	Terms of π	Approximate Value
	$C = 4\pi \text{ cm}$	Use $\pi = \frac{355}{113}$. $C = 12\frac{64}{113} \text{ cm}$
	$C = 2.2\pi \text{ cm}$	Use $\pi = 3.14$. $C = 6.908 \text{ cm}$
	$C = 4\frac{2}{3}\pi \text{ cm}$	Use $\pi = \frac{22}{7}$. $C = 14\frac{2}{3} \text{ cm}$
	$C = 2\frac{1}{2}\pi \text{ cm}$	Use $\pi = 3.14$. $C = 7.85 \text{ cm}$



4.6.3 Guided Practice: Error Analysis

1. Delanie used $\pi = \frac{22}{7}$ to find the circumference of the circle shown.



Delanie's work is shown.

Delanie's Work

$$\begin{aligned} C &= \pi d \\ C &= \frac{22}{7} \left(21 \frac{6}{7} \right) \\ C &= \frac{22}{7} \left(\frac{153}{7} \right) \\ C &= \frac{3366}{7} \\ C &= 480 \frac{6}{7} \text{ m} \end{aligned}$$

- a. Examine Delanie's work and write an explanation to her of any errors.

Delanie substituted the radius instead of the diameter and did not multiply the denominators, which resulted in an incorrect answer.

- b. What is the circumference of the circle?

$$C = 137 \frac{19}{49} \text{ m}$$

4.6.3 GUIDED PRACTICE: ERROR ANALYSIS

ENRICHMENT

What does $21 \frac{6}{7}$ meters represent? What formula did Delanie use? What substitutions did Delanie make?

TEACHER MOVES

Students may also notice that Delanie did not correctly multiply the denominators in her work.

4.6.4 YOUR TURN!

TEACHER MOVES

The phrase “linear fencing” in question 1 may confuse students. Refer the student to the picture and ask the student, if the fence was sold in a store, would it be circular or in a straight line? Also, if the student connects the “unpeeling” of the circumference in the exploration in Lesson 5, it may help them to understand why the question asks for linear feet.

Students may wonder what form of pi to use to answer question 2. In this case, it is best to use 3.14 or the exact value of pi, and then round to the desired place value.



4.6.4 Your Turn!

- Stephanie is putting a fence around a circular garden.

If the garden has a diameter of 12.5 feet (ft), how many linear feet (ft) of fencing does Stephanie need to purchase? Round your answer to the nearest foot.



$$C = \pi d = 3.14 \times 12.5\text{ft} = 39.25 \approx 40\text{ ft}$$

Stephanie should purchase 40 ft of fencing to make sure she has enough.

- Rachel needs a new tube for her bicycle tire. The tire has a radius of 13 inches (in.). What is the circumference of the tire tube that Rachel should buy?



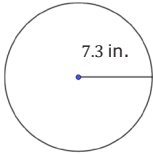
$$C = 2\pi r = 2 \times 3.14 \times 13\text{in} = 81.64\text{ inches}$$

- Natalia noticed that her hula-hoop is smaller than Felix's.
 - If Felix's hula-hoop has a diameter of 28 in., what is the circumference of his hula-hoop?
 $C = d\pi = 28\text{in} \times 3.14 = 87.92\text{ in}$
 - Natalia knows that her hula-hoop has a circumference of 81.64 in. Approximately how much larger is the circumference of Felix's hula-hoop than Natalia's?
 $d = 87.92 - 81.64 = 6.28\text{ in}$



4.6.5 Wrap-Up: Ticket Out the Door

Aziyia and Jamal are both finding the circumference of the circle shown. The radius of the circle is 7.3 inches (in.).



Some of Aziyia’s and Jamal’s work is shown.

Aziyia’s Work	Jamal’s Work
$C = 2\pi r$	$C = \pi d$
$C = (2)(3.14)r$	$C = \pi(2 \cdot 7.3)$
$C = (2)(3.14)(7.3)$	$C = 14.6\pi$ in.
$C = 6.28(7.3)$	
$C = 45.844$ in.	

- 1. Complete each person’s work.
See above.
- 2. Explain which answer is more accurate.
Jamal’s answer is more accurate because it is in terms of π rather than an approximation.

4.6.5 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

MA.K12.MTR.3.1: Mathematical Fluency

Fluency in this activity means flexibility in problem-solving. Students are identifying and completing two different methods for finding circumference; in one, they are using the radius, and in another, they are using the diameter (by multiplying the radius by two).